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Final Case Study

A Frontend Development Delima

A frontend developer's job is to design an interactive interface that makes tasks easier and more logical to complete. In industry, there are two main types of applications frontend developers are asked to design. Internal applications are created to help a company track and manage their own assets. External applications are created to allow customers to access products and services a company may offer. Depending on the purpose and customer criteria an application may have, there are certain design choices a front-end developer may make. For instance, internal applications may be more bare bones while external applications typically offer more aesthetic designs to give customers the best possible experience. Some external applications even use bold colors, enlarged text, and popups to try to coerce customers into purchasing subscriptions or other products. So, although frontend applications may not have an immediately obvious impact on the well-being of the public, there are small design decisions that can impact those who use the app.

You are a frontend developer working for an online shopping (e-commerce) company. You are the lead app developer working with your long-time friends and co-workers Joe and Mary. As part of your compensation, you and your colleagues get a bonus for every order made through the app (the bigger the order, the more compensation you get). Your boss, Randy, wants to try to maximize orders by making the "Add to cart" button enlarged, hiding the "Remove from cart" button under a small icon, and automatically selecting the "Add warranty" button for all eligible items. Although these design decisions comply with company rules and local laws, you are torn on whether adding these changes is ethical.

There are several stakeholders involved in this case. The first stakeholder is yourself. You are a hardworking individual with a big heart. You don't want to harm anyone and want to leave a positive impact on the world. You also want to make more money to help provide for your family and would like to see your friends thriving by getting a large bonus as well. The next stakeholders are Joe and Mary. Joe and Mary can add suggestions to the application design but ultimately have no say in the final design. Joe and Mary are both excited about the potential bonuses to supplement their income. Randy, one of the company executives, is a stakeholder. Randy, with your advice, is the one responsible for ensuring the production of a high-quality application. He is under a lot of pressure to

maximize profits as the company didn't hit its revenue mark last year. He doesn't have any programming experience and passes his ideas to you to develop with your co-workers. The company is also a stakeholder. They want to make a large profit but also don't want to ruin their public perception as a top-tier e-commerce company. The future consumers are the final stakeholders. They want an app that is intuitive and can help them complete their online shopping faster. The users of the application will most likely be from a large range of backgrounds. Some are poor, looking for cheap deals to save money. Others are rich, looking for ensured high quality products.

Although you may not have the final design decision, there are still several different options that you have. One option is to listen to Randy's wishes and add all three designs he wanted to implement. Another option is to develop some of the things Randy wants but discuss your discomfort with him. You could refuse to implement the changes at all costs and step away from the company. You could give Joe and Mary separate instructions and design things behind Randy's back. Or you could be upfront with Randy and suggest a different design knowing that Randy still has the final say. Given the wide range of options you have, it is best to use some of the ethical principles covered in this class to evaluate each individual issue and option.

As discussed in my term paper, to make strong ethical decisions, it is important to view each decision from multiple perspectives. That is why I am applying multiple frameworks to this case. The first framework I will apply is Deontology. I want to look at the specific rules that the company has laid out and make sure I am not violating any in the decision I make. Additionally, I want to look from an Agent-centered perspective to see how a decision directly affects myself and my peers. This is relevant to my case because the company's rules are sort of laid out and may have an impact on the decision-making process. I will also apply Consequentialism. I want to explore the likely consequences of a decision. I want to see how each decision affects each stakeholder differently. Ultimately, the best outcome for each stakeholder might be the decision I end up making. This is relevant to my case because each potential decision will have a wide variety of consequences. It is important to weigh the potential outcomes. The final framework I will apply is Virtue-Based Ethics. I want to specifically look at how respect and determination may play a role in the decision-making process. How can I respect each stakeholder and the world around me? How can I be determined to be the best engineer I can be? This is relevant because one of the main deciding factors of a potential decision has to do with the virtues held by the one making the decision.

Looking at this case from a deontological point of view, the option to follow Randy's wishes seems to be a very good option. Not only are you not breaking any company rules or

laws, but you are also benefiting yourself and your friends Joe and Mary by increasing the potential bonuses which is important in agent-centered deontology. Similarly, implementing some of Randy's suggestions and expressing your concern would be a good option from a deontological point of view. Refusing to work on the project and stepping away from the company doesn't break any company rules or laws, but from an agent-centered view, putting yourself out of a job would be disadvantageous and even potentially harmful for those reliant on your income. Going behind Randy's back and telling Joe and Mary to implement a different design would likely violate company rules, especially because you were not given the final say on the design. Therefore, this would not be a very ethical decision from a deontological point of view. And finally, being upfront with Randy suggesting a new design wouldn't break any rules but would likely limit the bonuses you and your co-workers would receive. Therefore, this option wouldn't be the best from a deontological point of view but also wouldn't be too bad. So, from a deontological perspective, option 1 and 2 are the best, option 5 is okay, and options 3 and 4 are the worst.

Switching to a consequentialist point of view, following Randy's wishes seems less enticing. Multiple stakeholders would be harmed in this scenario. First, app users would be potentially tricked into accidental purchases which could really hurt them if they are low on money. Second, the company would probably get some negative backlash from angry customers who made these accidental purchases. Third, Randy may face consequences for suggesting this design if the company receives enough backlash. However, you and your co-workers would still get the bonuses from the additional purchases. Only implementing some of Randy's design and confronting him about your concerns would be a slightly better option. However, Randy's design was created in attempt to deceive customers so even implementing part of it would contribute to the same consequences. Refusing to implement the design and leaving would also result in a similar outcome because the company would likely just hire someone else to do your job. Except for this case, you would be out of a job and separated from your friends Joe and Mary, making it even worse. Giving Joe and Mary instructions to implement a design that is less deceiving would be a much better option from a consequentialist view. App users would be able to use the app without being deceived. The company wouldn't receive complaints about the deception. Randy would be praised for a solid app design. However, you would potentially get into trouble for going behind Randy's back. You and your co-workers would also get less bonus money. However, the best case from a consequentialist point of view would be to directly talk to Randy about your concerns. From there, you could convince Randy to take a more ethical approach in his design, and it would result in similar consequences as going behind Randy's except you would likely receive praise from Randy for your suggestions instead of face consequences for going behind his back. Keep in mind, consequentialism states that

only the most ethical decision is ethical. Therefore, even though going behind Randy's back is the second-best option from this view, it is technically unethical. So, from a consequentialist point of view, option 5 is ethical and options 1, 2, 3, and 4 are unethical.

Finally, looking from a virtue-based perspective with an emphasis on respect and determination, following Randy's wishes shows respect for Randy and his judgement. However, it shows a lack of respect for the future users of the app who may be deceived into making unnecessary purchases. Additionally, there is a lack of determination by yourself to do the right thing. Again, implementing most of Randy's suggestions and talking to him about why you don't feel comfortable doing all of them has a similar impact on virtue-based ethics. The only reason why this option is slightly better is because you show more determination in trying to do the right thing. Completely refusing to do implement Randy's design and leaving the company is one of the worst decisions from a virtue-based perspective. You show no respect for Randy by failing to acknowledge him and explaining why his design might not be the best. You show no respect for the future consumers of the app because you leaving just means someone else will do it. You show no respect for yourself because you are quitting a job. This also ties to determination; by choosing this option you show no determination to do the right thing and just quit because it is too hard to stand up against something that is potentially wrong. Going behind Randy's back to complete the project in a non-deceitful way shows respect for the future consumers, but completely disrespects Randy's ideas. It also disrespects Joe and Mary because they think they are creating the application design that the company wants, but they are really creating your design. Finally, you show some determination to do the right thing by choosing this option, but you fail to do it in the right way. Avoiding Randy is the easiest way to create a completely ethical app, but it isn't the right way to do it. The best option from a virtue-based perspective is to talk to Randy to try to convince him to do the right thing in his design. Doing so shows respect for the future app users, Randy, Joe, and Mary, all while maintaining your determination to do the right thing. So, from a virtue-based perspective option 5 is the best, while options 1, 2, and 4 show some weaknesses, and option 3 is the worst.

By reviewing all three ethical frameworks, the best decision becomes quite clear. What you should do in this situation is talk to Randy and explain why you are uncomfortable with his design. You can explain to him why you understand his need to improve profits and that although the techniques aren't terribly uncommon in the e-commerce industry, they aren't necessarily ethical. Additionally, you can explain that if he chooses a more intuitive design, there will be higher user retention and less chance of complaints being sent to the company. Through this decision, you will hopefully be able to convince Randy to create a more ethical design that you and your team can confidently

build. This decision seems to be the most ethical because by applying the three ethical principles we could break down the complex implications of this scenario. This option was the best option in both consequentialism and virtue-based ethics and was one of the better options from a deontological perspective. So no, it wasn't a unanimous decision, but often with ethics, the ethical principles are going to somewhat contradict each other. The best thing to do is look at each separate framework and then weigh them altogether. Through this process, we found that talking to Randy was the best option.